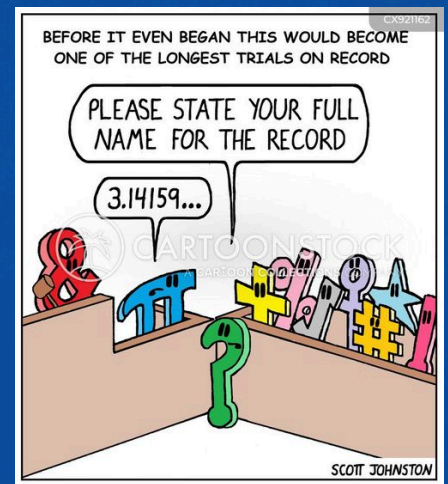




Student Name: HEWP THOS LOPF TRIA AJIA CHEY

Test Name: MATHEMATICAL METHODS

Date: _____



MATHEMATICS TEST

UNIT 2

Test theme: EXAM TECH FREE

Reading time: 10 minutes

Writing time: 60 minutes

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
10	10	50

- Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers.
- Students are NOT permitted to bring into the examination: notes of any kind, blank sheets of paper, white out liquid/tape or a calculator of any type.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your name in the space provided above on this page.
- All written responses must be in English or Romanian.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Instructions

Answer **all** questions in the spaces provided.

Unless otherwise specified an **exact** answer is required to a question.

In questions where more than one mark is available, appropriate working **must** be shown.

Question 1 (5 marks)

Solve the following for x :

a. $\log_2(x) = \log_4(2x + 3)$

3 marks

b. $2^{x+2} + 2^{1-x} = 9$

2 marks

Question 2 (4 Marks)

Consider the function: $f: D \rightarrow R, f(x) = \sqrt{x^2 - 4x - 5}$

- a. Determine the maximal domain D of $f(x)$.

2 Marks

- b. Find $f'(x)$ and state the domain for which $f'(x)$ is defined

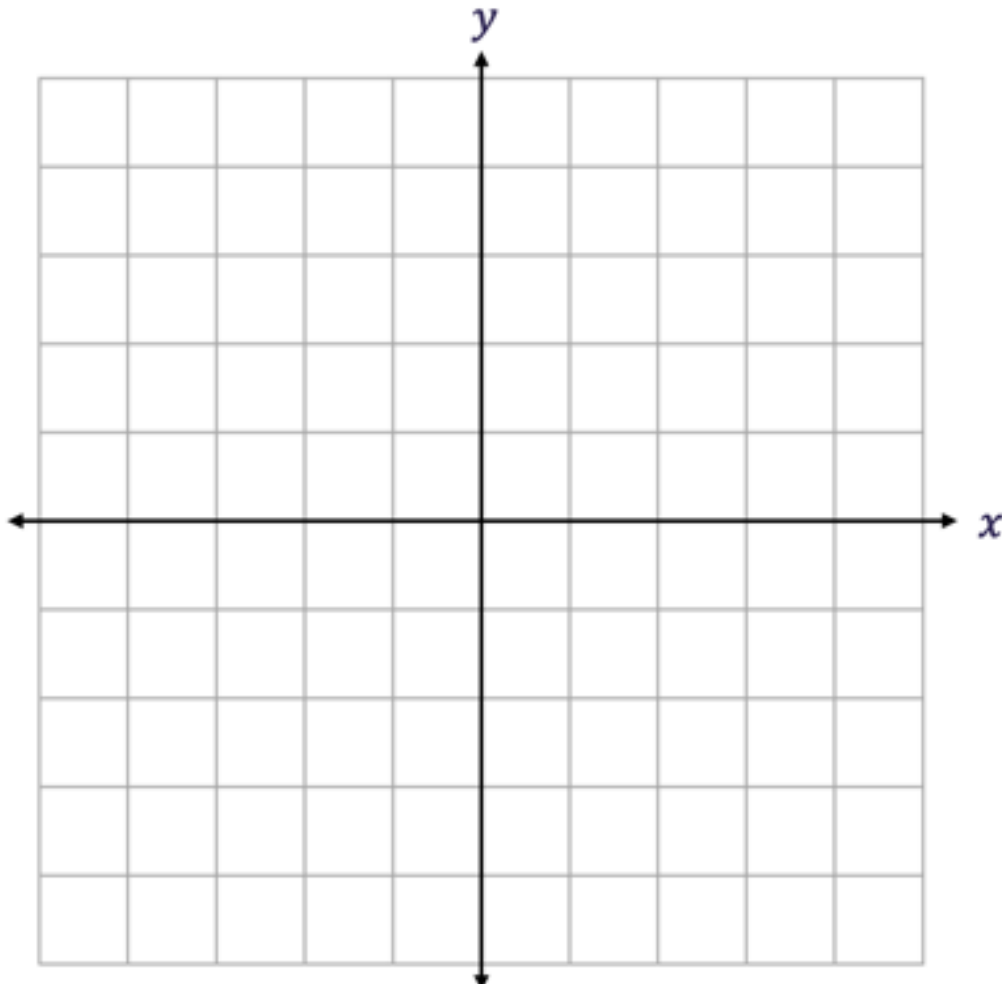
2 Marks

Question 3 (5 Marks)

Consider the function $g(x) = x^2 - 6x + 5$

- a. Sketch the graph of $y = g(x)$, labelling the stationary points and all axial intercepts

2 Marks



- b. On the same set of axes, sketch the graph of $h(x) = \sqrt{g(x)}$. Label the coordinates of the invariant points and any endpoints

3 Marks

Question 4 (4 Marks)

Use the functional derivative formula (differentiation from first principles) to find the derivative of $f(x) = \sqrt{x + 3}$.

Question 5 (6 Marks)

A curve is defined by the equation $y = \frac{1}{x^2}$.

- a. Find the equation of the tangent to the curve at the point where $x = 1$.

2 Marks

- b. This tangent intersects the x -axis at point A and the y -axis at point B . Find the coordinates of A and B .

2 Marks

- c. Calculate the area of the triangle OAB , where O is the origin.

2 Marks

Question 6 (5 Marks)

Solve the equation $\sin(2x - \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$ for $x \in [-\pi, \pi]$.

Question 7 (6 Marks)

A particle moves along a straight line such that its velocity, v (m/s) at time t (seconds) is given by $v(t) = 3t^2 - 12t + 9$ for $t \geq 0$

Using $\int v(t)dt = a(t)$, where $\int ax^n dx = a \times \frac{x^{n+1}}{n+1} + c$, where c is a constant. (c in this case is 0)

- a. Find the particle's displacement from its starting position at $t = 3$.

2 Marks

- b. Find the total distance travelled by the particle in the first 3 seconds of motion.

4 Marks

Question 8 (8 Marks)

An open-topped box is to be made from a rectangular piece of cardboard measuring 16cm by 10cm by cutting out a square of side length $x\text{cm}$ from each corner and folding up the sides.

- a. Show that the volume $V(x)$ of the box is given by $V(x) = 4x^3 - 52x^2 + 160x$

2 Marks

- b. State the restricted domain of x for which this box can be physically constructed.

1 Mark

- c. Find the value of x that maximises the box's volume.

3 Marks

- d. Calculate the maximum volume.

2 Marks

Question 9 (4 Marks)

Find the coordinates of the point on the curve $y = 2x^2$ that is closest to the point (0, 5).

4 Marks

Question 10 (3 Marks)

Consider the function $f(x) = x^3 - 3x^2 + k$, where $k \in \mathbb{Z}$.

Determine the value(s) of k such that the graph $y = f(x)$ has exactly two x -intercepts.

3 Marks

END OF EXAM